

T.C.

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THE EFFECT OF AMOUNT OF VAT COLLECTED FROM ELECTRONICS ON AMOUNT OF PLASTICS IN WATER FROM ELECTRONICS IN TURKEY

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1. INTRODUCTION

The goal of this study is to thoroughly examine a specific problem area. The process of developing these models is given special consideration, starting with the identification of the problem area. To ensure a focused and accurate analysis, this involves defining the parameters of the study and drawing precise boundaries around the systems. The Causal Loop Diagram (CLD) is an essential tool for understanding and visualizing the feedback mechanisms that are built into the system under study. It shows the interactions between different variables. The focus then shifts to using the STELLA software to create a stock flow diagram, where the first step is to define and identify stocks. The basis for the development of interactions between these variables is then established by identifying flows, converters and additional variables. These interactions are further defined and arranged through the development of mathematical equations with precise units. To populate the model with accurate data, a comprehensive data collection technique is employed to acquire information from a variety of databases, research papers and publications. A thorough unit consistency check is performed once the data has been integrated into the model to guarantee its coherence and reliability. The next phase comprises a comprehensive verification and validation process to ensure that the model is accurate and suitable for solving the problem in consideration. In the final phase of the study, tables and graphs are included to improve the visual perception of the model and the understanding of the interaction between variables. Furthermore, the sensitivity analysis provides insights by discussing the behavior and results of the system under different parameters.

This study demonstrates the effectiveness of system dynamics modeling in understanding and forecasting the behavior of complex systems. The model's ability to predict future system behavior, its adaptability to real world scenarios and its potential to provide policymakers with strategic information to support their decision making are all significant advantages of this technique. This approach might help us better understand the dynamics of complex systems and provide solutions to the issues they encounter.

2. PROBLEM STATEMENT

The project's aims are to understand how the amount of Value Added Tax (VAT) collected from electronics influences the amount of plastic in water derived from electronic devices. Along with the increase in electronic device sales, the total amount of VAT collected changes, which in turn affects both the usage habits of electronic devices in Turkey and the generation of electronic waste.

Changes in the VAT rates applied to electronic products alter the amount of tax collected by the government as well as the budget allocated from these revenues to waste management. The size of the funds set aside for waste management leads to significant changes in the expenditures dedicated to recycling and removal processes. An increase in the recycling rate facilitates the recovery of collected waste and its conversion to secondary use.

Especially in institutions such as hospitals, products reused for secondary purposes eventually reach the end of their service life and return to the waste stream. Through this cycle, the recycling process contributes both to reducing overall waste and, at certain stages, to increasing the amount of waste resulting from recycling.

This analysis covers the relationship between VAT rate changes, the funds allocated to waste management, recycling rates and the amount of plastic in water bodies. The findings aim to provide recommendations for developing environmentally friendly policies.

3. ASSUMPTIONS

In this study, we ran a 20 year simulation (from 2024 to 2044) to capture the long term interactions among electronic product supply, government funding and amount of plastics in Turkish waters. All monetary transactions are expressed in US dollars for ease of comparison, using a fixed exchange rate of 1 USD = 23.9 TL, based on the 2023 Medium Term Program. No adjustments for inflation or future exchange rate fluctuations are included, allowing us to focus on core relationships rather than external economic changes.

We represent Turkey's entire electronics market with stocks and assume that all devices share the same average product lifespan of 6 years and contain a standard average plastic content (about 0.88625 kilograms or 886.25 grams per device). 886.25 grams may seem quite heavy considering the electronics commonly used in daily life, but since we took all the electronics into consideration when designing the model, 886.25 g is a reasonable value when averaged. Once devices reach end of life, they can enter multiple pathways: a portion is recycled (recycling rate varies by expenditure on recycling programs). Some of the recycled plastics are also reprocessed for use as medical equipment at hospitals. And the rest ends up as plastic waste, from which the specific rate (plastic removal rate varies by expenditure on plastic removal) is removed by dedicated government programs and the remainder slowly degrades in water over an average decomposition time of 300 years. We also assume any plastics reprocessed at hospitals as medical equipments eventually return to the waste stream as End of Life Materials.

On the policy side, VAT Collected from Electronics and All Other Collected Taxes flow into a Total Government Budget, a fixed fraction (which is 1.5% - this rate was estimated by taking into consideration the budget allocations of statistical institutions in Turkey and around the world) of which is allocated to waste management each year. This fund is then split among recycling programs, plastic removal and administrative affairs, with each category assumed to have a direct impact on the corresponding rates.

The ceteris paribus (all else equal or all other things being equal) approach applies when analyzing economic effects. For the sake of simplicity, All Other Collected Tax is assumed to be constant at 177.62 Billion USD, although this may seem unrealistic given GDP growth, the only tax revenue that changes annually is the Amount of VAT Collected from Electronics.

Another critical assumption is that the price elasticity of electronics around the market price is between 0 and 1 (inelastic) and that the price elasticity varies by movement along the price-response curve. We incorporated this assumption into our model with the Effect of Price on Sales converter graph. This assumption provides that: when the price of electronics rises by a given percentage, the sales of electronics fall by a lower percentage and the government's tax revenue rises overall. Vice versa, when the price of electronics falls, the sales of electronics rise at a lower percentage and the government's tax revenue falls. Considering microeconomics and pricing theories and models, this assumption is not quite reasonable. Realistic logit price-response graphs are reserve s-shaped or sigmoid. Willingness to pay graph would be logistic distribution similar to bell shape but with flatter tails. Our price-response graph is also close to realistic but not as logit. Thus, sales of electronics does not provide very realistic values, but since this assumption is taken into consideration even in microeconomics and pricing theories for simplicity, we considered it reasonable to use this assumption in our model.

In terms of workforce, our model treats the electronics industry’s hiring and firing decisions as primarily motivated by differences between desired production output (influenced by inventory levels) and the existing number of workers. Lastly, we do not implement any design improvement measures in this project phase, keeping technology and process parameters constant throughout the 20 year horizon.

4. CAUSAL LOOP DIAGRAM

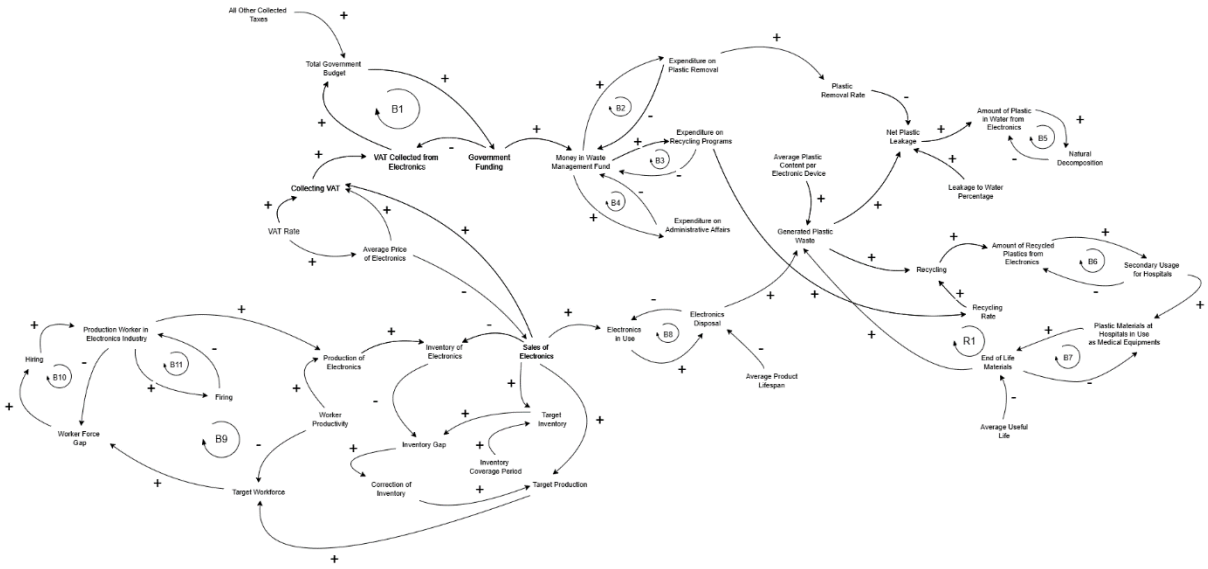


Figure 4.1. Causal Loop Diagram (CLD)

The causal loop diagram (CLD) maps the interconnections between economic, environmental and industrial factors, focusing on the effect of Amount of VAT Collected from Electronics on Amount of Plastics in Water from Electronics in Turkey. The system represents the system dynamics underlying the interactions between the taxation of electronics, the supply chain of the electronics industry, the government budgeting, the management of electronic waste and their environmental impact on water. Here's a detailed breakdown:

Paths, Types and Explanations of the Feedback Loops:

1. **Path:** Total Government Budget — Government Funding — VAT Collected from Electronics (B1)

   Type: Balancing Loop

   Explanation: As VAT Collected from Electronics increases, Total Government Budget increases and vice versa, as VAT Collected from Electronics decreases, Total

Government Budget decreases. That is, there is a same way interaction between them. As Total Government Budget increases, Government Funding increases. This interaction represents that as the government budget increases, it will allocate more money for waste management or other funding. As Government Funding increases, VAT Collected from Electronics decreases. The reason for this decrease is that Government Funding is an outflow of VAT Collected from Electronics, which is a cumulatively accumulated stock. The feedback mechanism as a whole is a balancing loop and seeks to stabilize the system.

2. **Path:** Money in Waste Management Fund — Expenditure on Plastic Removal (B2)
   Type: Balancing Loop
   Explanation: As Money in the Waste Management Fund increases, Expenditure on Plastic Removal also increases. This interaction indicates that the government could allocate more resources for plastic removal as the Money in Waste Management Fund increases. As Expenditure on Plastic Removal increases, Money in the Waste Management Fund decreases. The feedback mechanism as a whole is a balancing loop and seeks to stabilize the system.

3. **Path:** Money in Waste Management Fund — Expenditure on Recycling Programs (B3)
   Type: Balancing Loop
   Explanation: As Money in the Waste Management Fund increases, Expenditure on Recycling Programs also increases. This interaction indicates that the government could allocate more resources for recycling programs as Money in Waste Management Fund increases. As Expenditure on Recycling Programs increases, Money in the Waste Management Fund decreases. The feedback mechanism as a whole is a balancing loop and seeks to stabilize the system.

4. **Path:** Money in Waste Management Fund — Expenditure on Administrative Affairs (B4)

   Type: Balancing Loop

   Explanation: As Money in Waste Management Fund increases, Expenditure on Administrative Affairs also increases. This interaction indicates that the government could allocate more resources for administrative affairs as Money in Waste Management Fund increases. As Expenditure on Administrative Affairs increases, Money in the Waste Management Fund decreases. The feedback mechanism as a whole is a balancing loop and seeks to stabilize the system.

5. **Path:** Amount of Plastics in Water from Electronics — Natural Decomposition (B5)

   Type: Balancing Loop

   Explanation: As Amount of Plastics in Water from Electronics increases, Natural Decomposition also increases. As Natural Decomposition increases, Amount of Plastics in Water from Electronics decreases. The feedback mechanism as a whole is a balancing loop and seeks to stabilize the system. It is also important to mention that the decreases are quite slow here as Natural Decomposition takes a very long time.

6. **Path:** Amount of Recycled Plastics from Electronics — Secondary Usage for Hospitals (B6)

   Type: Balancing Loop

   Explanation: As Amount of Recycled Plastics from Electronics increases, Secondary Usage for Hospitals also increases. As Secondary Usage for Hospitals increases, Amount of Recycled Plastics from Electronics decreases. The feedback mechanism as a whole is a balancing loop and seeks to stabilize the system.

7. **Path:** Plastic Materials at Hospitals in Use as Medical Equipments — End of Life Materials (B7)

   Type: Balancing Loop

   Explanation: As Plastic Materials at Hospitals in Use as Medical Equipments increases, End of Life Materials also increases. As End of Life Materials increases, Plastic Materials at Hospitals in Use as Medical Equipments decreases. The feedback mechanism as a whole is a balancing loop and seeks to stabilize the system.

8. **Path:** Generated Plastic Waste — Recycling — Amount of Recycled Plastics from Electronics — Secondary Usage for Hospitals — Plastic Materials at Hospitals in Use Medical Equipments — End of Life Materials (R1)

   Type: Reinforcing Loop

   Explanation: As Generated Plastic Waste increases, Recycling also increases. As Recycling increases, Amount of Recycled Plastics from Electronics also increases. As Amount of Recycled Plastics from Electronics increases, Secondary Usage for Hospitals also increases. As Secondary Usage for Hospitals increases, Plastic Materials at Hospitals in Use Medical Equipments also increases. As Plastic Materials at Hospitals in Use Medical Equipments increases, End of Life Materials also increases. As End of Life Materials increases, Generated Plastic Waste also increases. Since all interactions are the same way, the loop is reinforcing. As the loop is reinforcing, unlike other feedback mechanisms in the system, waste in this loop increases by acceleration due to the loop structure.

9. **Path:** Electronics in Use — Electronics Disposal (B8)

   Type: Balancing Loop

   Explanation: As Electronics in Use increases, Electronics Disposal also increases. As Electronics Disposal increases, Electronics in Use decreases. The feedback mechanism as a whole is a balancing loop and seeks to stabilize the system.

10. **Path:** Inventory of Electronics — Inventory Gap — Correction of Inventory — Target Production — Target Workforce — Workforce Gap — Hiring — Production Worker in Electronics Industry — Production of Electronics (B9)

    Type: Balancing Loop

    Explanation: As Inventory of Electronics increases, Inventory Gap decreases. As Inventory Gap decreases, Correction of Inventory also decreases. As Correction of Inventory decreases, Target Production also decreases. As Target Production decreases, Target Workforce also decreases. As Target Workforce decreases, Workforce Gap also decreases. As Workforce Gap decreases, Hiring also decreases. As Hiring decreases, Production Worker in Electronics Industry also decreases. As Production Worker in Electronics Industry decreases, Production of Electronics also decreases. As Production of Electronics decreases, Inventory of Electronics also decreases. The feedback mechanism as a whole is a balancing loop and seeks to stabilize the system.

11. **Path:** Production Worker in Electronics Industry — Firing (B10)

    Type: Balancing Loop

    Explanation: As Production Worker in Electronics Industry increases, Firing also increases. As Firing increases, Production Worker in Electronics Industry decreases. The feedback mechanism as a whole is a balancing loop and seeks to stabilize the system.

12. **Path:** Production Worker in Electronics Industry — Workforce Gap — Hiring (B11)

    Type: Balancing Loop

    Explanation: As Production Worker in Electronics Industry increases, Workforce Gap decreases. As Workforce Gap decreases, Hiring also decreases. As Hiring decreases, Production Worker in Electronics Industry also decreases. The feedback mechanism as a whole is a balancing loop and seeks to stabilize the system.

5. DYNAMIC MODEL

5.1. STOCK-FLOW DIAGRAM

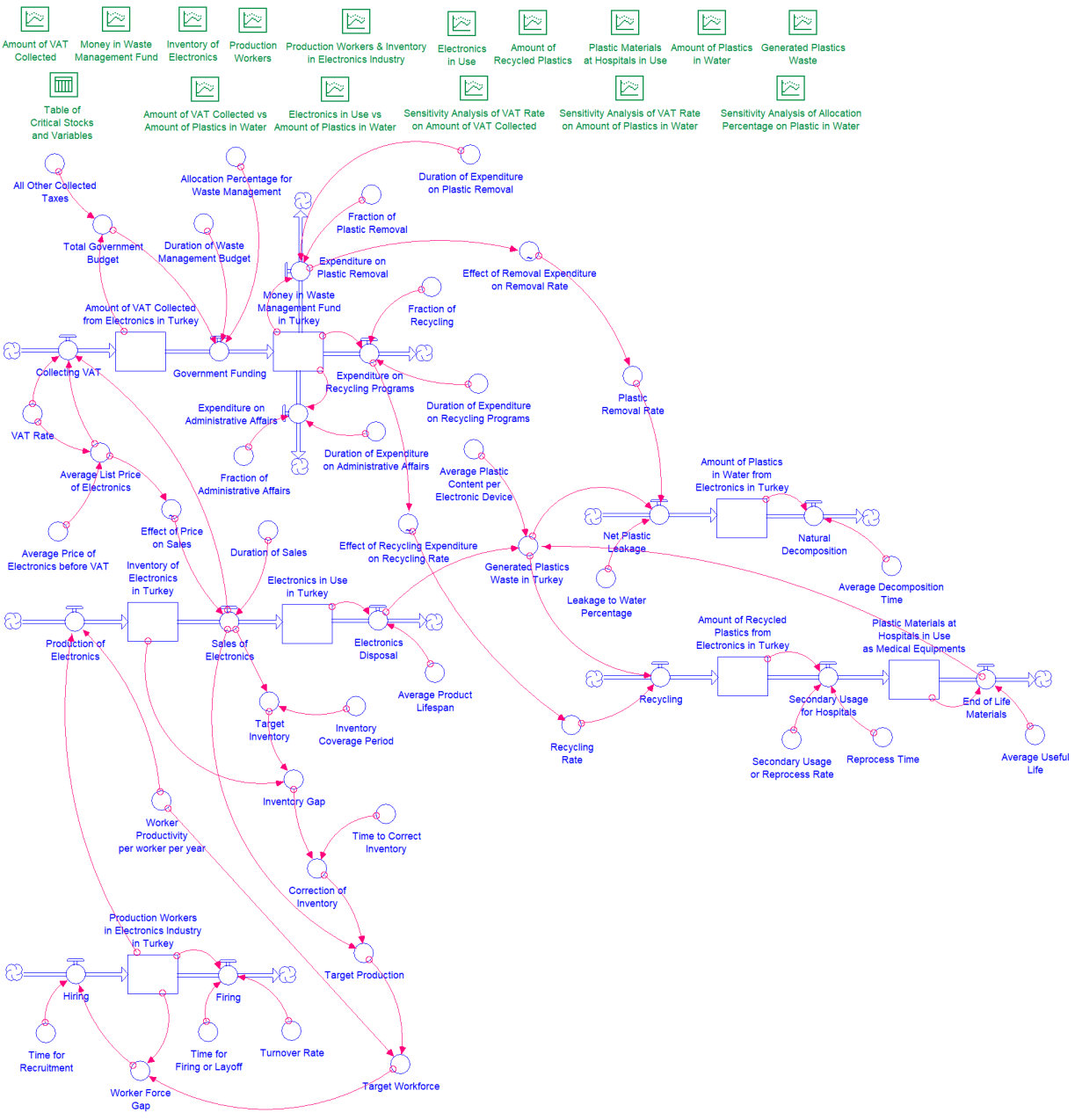


Figure 5.1. Stock-Flow Diagram

The stock-flow diagram offers a holistic, system level view of how electronic products, plastic waste, government budgets and labor dynamics interact over time. In this structure, the primary stocks (illustrated as rectangular nodes) represent accumulations such as Inventory of Electronics, Production Workers in the Electronics Industry, Money in the Waste Management Fund and the Amount of Plastics in Water from Electronics. Each stock changes according to its respective inflows and outflows (shown as rate variables), capturing how materials, financial resources and workforce numbers evolve within the system.

From an industrial perspective, Production of Electronics depends on factors such as Target Workforce, Hiring, Firing and Inventory Gap. Firms adjust production and staffing to maintain an optimal inventory level and meet market demand. Workforce additions occur when a Workforce Gap emerges, driving Hiring rates upward; conversely, an excess of workers elevates Firing rates. These balancing relationships stabilize the number of workers, production levels and ultimately the Inventory of Electronics.

On the environmental front, once electronics are produced and eventually disposed of, plastics enter the waste stream. The diagram tracks this progression through a network of flows (Generated Plastic Waste, Recycling and eventual Amount of Plastics in Water from Electronics). This last stock depletes slowly via Natural Decomposition and is also reduced through Expenditure on Plastic Removal programs. Meanwhile, Recycled Plastics can reenter the economy for secondary uses such as Plastic Materials at Hospitals in Use as Medical Equipments, then ultimately progress toward End of Life Materials, feeding back into waste streams.

The government budget structure links fiscal mechanisms to waste management and recycling activities. As VAT Collected from Electronics and other revenue streams flow into the Total Government Budget, allocations to the Waste Management Fund support expenditures on Administrative Affairs, Recycling Programs and Plastic Removal. Each of these outflows affects the fund’s balance, shaping the capacity for subsequent waste related initiatives and programs.

Overall, the diagram underscores the interconnectedness of economic, social and environmental processes. By representing stocks as accumulations and flows as transfers or transformations, it vividly illustrates how policies, resource availability, workforce adjustments and material life cycles influence one another over time. This structure lays the groundwork for analyzing policy scenarios, identifying leverage points and fostering sustainable strategies in electronics production and waste management.

5.2. UNITS

5.2.1. UNITS OF STOCKS

- Amount of VAT Collected from Electronics in Turkey: US Dollars
- Money in Waste Management Fund in Turkey: US Dollars
- Inventory of Electronics in Turkey: Electronic Device
- Electronics in Use in Turkey: Electronic Device
- Production Workers in Electronics Industry in Turkey: workers
- Amount of Recycled Plastics from Electronics in Turkey: kilograms
- Plastic Materials at Hospitals in Use as Medical Equipments in Turkey: kilograms
- Amount of Plastics in Water from Electronics in Turkey: kilograms

5.2.2. UNITS OF FLOWS

- Collecting VAT: US Dollars/years
- Government Funding: US Dollars/years
- Expenditure on Recycling Programs: US Dollars/years
- Expenditure on Plastic Removal: US Dollars/years
- Expenditure on Administrative Affairs: US Dollars/years
- Production of Electronics: Electronic Device/years

- Sales of Electronics: Electronic Device/years
- Electronics Disposal: Electronic Device/years
- Hiring: workers/years
- Firing: workers/years
- Net Plastic Leakage: kilograms/years
- Natural Decomposition: kilograms/years
- Recycling: kilograms/years
- Secondary Usage for Hospitals: kilograms/years
- End of Life Materials: kilograms/years

5.2.3. UNITS OF CONVERTERS

- All Other Collected Taxes: US Dollars
- Total Government Budget: US Dollars
- Effect of Price on Sales: Electronic Device
- Target Inventory: Electronic Device
- Inventory Gap: Electronic Device
- Worker Force Gap: workers
- Target Workforce: workers
- Duration of Waste Management Budget: years
- Duration of Expenditure on Plastic Removal: years
- Duration of Expenditure on Administrative Affairs: years
- Duration of Expenditure on Recycling Programs: years
- Average Decomposition Time: years
- Reprocess Time: years
- Average Useful Life: years
- Duration of Sales: years
- Inventory Coverage Period: years
- Average Product Lifespan: years

- Time to Correct Inventory: years
- Time for Recruitment: years
- Time for Firing or Layoff: years
- Average List Price of Electronics: US Dollars/Electronic Device
- Average Price of Electronics before VAT: US Dollars/Electronic Device
- Average Plastic Content per Electronic Device: kilograms/Electronic Device
- Generated Plastics Waste in Turkey: kilograms/years
- Correction of Inventory: Electronic Device/years
- Target Production: Electronic Device/years
- Worker Productivity per worker per year: Electronic Device/workers/years
- VAT Rate: Unitless
- Allocation Percentage for Waste Management: Unitless
- Fraction of Plastic Removal: Unitless
- Fraction of Administrative Affairs: Unitless
- Fraction of Recycling: Unitless
- Effect of Recycling Expenditure on Recycling Rate: Unitless
- Plastic Removal Rate: Unitless
- Leakage to Water Percentage: Unitless
- Recycling Rate: Unitless
- Secondary Usage or Reprocess Rate: Unitless
- Effect of Removal Expenditure on Removal Rate: Unitless
- Turnover Rate: Unitless

5.3. EQUATIONS

Amount_of_Plastics_in_Water_from__Electronics_in_Turkey(t) =
Amount_of_Plastics_in_Water_from__Electronics_in_Turkey(t - dt) + (Net_Plastic_Leakage - Natural_Decomposition) * dt
INIT Amount_of_Plastics_in_Water_from__Electronics_in_Turkey = 5000000
INFLOWS:
Net_Plastic_Leakage = Generated_Plastics_Waste_in_Turkey*Leakage_to_Water_Percentage*(1-Plastic__Removal_Rate)
OUTFLOWS:
Natural_Decomposition = Amount_of_Plastics_in_Water_from__Electronics_in_Turkey/Average_Decomposition__Time

Amount_of_Recycled_Plastics_from_Electronics_in_Turkey(t) =
Amount_of_Recycled_Plastics_from_Electronics_in_Turkey(t - dt) + (Recycling - Secondary_Usage__for_Hospitals) * dt
INIT Amount_of_Recycled_Plastics_from_Electronics_in_Turkey = 10000000
INFLOWS:
Recycling = Generated_Plastics_Waste_in_Turkey*Recycling__Rate
OUTFLOWS:
Secondary_Usage__for_Hospitals = Amount_of_Recycled_Plastics_from_Electronics_in_Turkey*Secondary_Usage_or_Reprocess_Rate/Reprocess_Time

Amount_of_VAT_Collected__from_Electronics_in_Turkey(t) =
Amount_of_VAT_Collected__from_Electronics_in_Turkey(t - dt) + (Collecting_VAT - Government_Funding) * dt
INIT Amount_of_VAT_Collected__from_Electronics_in_Turkey = 4540000000
INFLOWS:
Collecting_VAT = VAT_Rate*Average_List_Price_of_Electronics*Sales_of__Electronics
OUTFLOWS:
Government_Funding = Total_Government__Budget*Allocation_Percentage_for__Waste_Management/Duration_of_Waste_Management_Budget

Electronics_in_Use_in_Turkey(t) = Electronics_in_Use_in_Turkey(t - dt) + (Sales_of__Electronics - Electronics_Disposal) * dt
INIT Electronics_in_Use_in_Turkey = 90000000
INFLOWS:
Sales_of__Electronics = Effect_of_Price__on_Sales/Duration_of_Sales
OUTFLOWS:
Electronics_Disposal = Electronics_in_Use_in_Turkey/Average_Product_Lifespan

Figure 5.2. Equations of the Dynamic Model (Part Ⅰ)

Inventory_of__Electronics_in_Turkey(t) = Inventory_of__Electronics_in_Turkey(t - dt) + (Production_of__Electronics - Sales_of__Electronics) * dt
INIT Inventory_of__Electronics_in_Turkey = 9000000
INFLOWS:
Production_of__Electronics = Worker__Productivity__per_worker_per_year*Production_Workers__in_Electronics_Industry_in_Turkey
OUTFLOWS:
Sales_of__Electronics = Effect_of_Price__on_Sales/Duration_of_Sales

Money_in_Waste__Management_Fund_in_Turkey(t) = Money_in_Waste__Management_Fund_in_Turkey(t - dt) + (Government_Funding - Expenditure_on__Recycling_Programs - Expenditure_on__Plastic_Removal - Expenditure_on_Administrative_Affairs) * dt
INIT Money_in_Waste__Management_Fund_in_Turkey = 1000000000
INFLOWS:
Government_Funding = Total_Government__Budget*Allocation_Percentage_for__Waste_Management/Duration_of_Waste_Management_Budget
OUTFLOWS:
Expenditure_on__Recycling_Programs = (Fraction_of__Recycling*Money_in_Waste__Management_Fund_in_Turkey)/Duration_of_Expenditure_on_Recycling_Programs
Expenditure_on__Plastic_Removal = (Fraction_of_Plastic_Removal*Money_in_Waste__Management_Fund_in_Turkey)/Duration_of_Expenditure_on_Plastic_Removal
Expenditure_on_Administrative_Affairs = (Fraction_of_Administrative_Affairs*Money_in_Waste__Management_Fund_in_Turkey)/Duration_of_Expenditure_on_Administrative_Affairs

Plastic_Materials_at__Hospitals_in_Use_as_Medical_Equipments_in_T(t) = Plastic_Materials_at__Hospitals_in_Use_as_Medical_Equipments_in_T(t - dt) + (Secondary_Usage__for_Hospitals - End_of_Life_Materials) * dt
INIT Plastic_Materials_at__Hospitals_in_Use_as_Medical_Equipments_in_T = 500000
INFLOWS:
Secondary_Usage__for_Hospitals = Amount_of_Recycled_Plastics_from_Electronics_in_Turkey*Secondary_Usage_or_Reprocess_Rate/Reprocess_Time
OUTFLOWS:
End_of_Life_Materials = Plastic_Materials_at__Hospitals_in_Use_as_Medical_Equipments_in_T/Average_Useful_Life

Figure 5.3. Equations of the Dynamic Model (Part Ⅱ)

Production_Workers__in_Electronics_Industry_in_Turkey(t) =
Production_Workers__in_Electronics_Industry_in_Turkey(t - dt) + (Hiring - Firing) * dt
INIT Production_Workers__in_Electronics_Industry_in_Turkey = 150000
INFLOWS:
Hiring = Worker_Force__Gap/Time_for__Recruitment
OUTFLOWS:
Firing =
(Turnover_Rate*Production_Workers__in_Electronics_Industry_in_Turkey)/Time_for_Firing_or_
Layoff

- Allocation_Percentage_for__Waste_Management = 0.015
- All_Other_Collected__Taxes = 177620000000
- Average_Decomposition__Time = 300
- Average_List_Price_of_Electronics = Average_Price_of__Electronics_before_VAT*(1+VAT_Rate)
- Average_Plastic__Content_per__Electronic_Device = 0.88625
- Average_Price_of__Electronics_before_VAT = 650/1.20
- Average_Product_Lifespan = 6
- Average_Useful_Life = 2.5
- Correction_of_Inventory = Inventory_Gap/Time_to_Correct__Inventory
- Duration_of_Expenditure_on_Administrative_Affairs = 1
- Duration_of_Expenditure_on_Plastic_Removal = 1
- Duration_of_Expenditure_on_Recycling_Programs = 1
- Duration_of_Sales = 1
- Duration_of_Waste_Management_Budget = 1
- Fraction_of_Administrative_Affairs = 0.15
- Fraction_of_Plastic_Removal = 0.25
- Fraction_of__Recycling = 0.60
- Generated_Plastics_Waste_in_Turkey =
  Electronics_Disposal*Average_Plastic__Content_per__Electronic_Device + End_of_Life_Materials
- Inventory_Gap = max(0, Target__Inventory-Inventory_of__Electronics_in_Turkey)
- Inventory__Coverage_Period = 3/12
- Leakage_to_Water_Percentage = 0.035
- Plastic__Removal_Rate = Effect_of_Removal_Expenditure_on_Removal_Rate
- Recycling__Rate = Effect_of_Recycling_Expenditure_on_Recycling_Rate
- Reprocess_Time = 1
- Secondary_Usage_or_Reprocess_Rate = 0.03
- Target_Production = Sales_of__Electronics+Correction_of_Inventory
- Target_Workforce = Target_Production/Worker__Productivity__per_worker_per_year
- Target__Inventory = Sales_of__Electronics*Inventory__Coverage_Period
- Time_for_Firing_or_Layoff = 60/12
- Time_for__Recruitment = 10/12

Figure 5.4. Equations of the Dynamic Model (Part Ⅲ)

Time_to_Correct__Inventory = 2/12

Total_Government__Budget = Amount_of_VAT_Collected__from_Electronics_in_Turkey+All_Other_Collected__Taxes

Turnover_Rate = 0.15

VAT_Rate = 0.20

Worker_Force__Gap = max(0, Target_Workforce-Production_Workers__in_Electronics_Industry_in_Turkey)

Worker__Productivity__per_worker_per_year = 230

Effect_of_Price__on_Sales = GRAPH(Average_List_Price_of_Electronics)



(100, 2.2e+008), (150, 1.5e+008), (200, 1.1e+008), (250, 8.9e+007), (300, 7.4e+007), (350, 6.4e+007), (400, 5.6e+007), (450, 5e+007), (500, 4.5e+007), (550, 4.1e+007), (600, 3.7e+007), (650, 3.4e+007), (700, 3.2e+007), (750, 3e+007), (800, 2.8e+007), (850, 2.6e+007), (900, 2.5e+007), (950, 2.4e+007), (1000, 2.2e+007)

Effect_of_Recycling_Expenditure_on_Recycling_Rate = GRAPH(Expenditure_on__Recycling_Programs)



(0.00, 0.00), (5e+007, 0.025), (1e+008, 0.05), (1.5e+008, 0.075), (2e+008, 0.1), (2.5e+008, 0.12), (3e+008, 0.14), (3.5e+008, 0.155), (4e+008, 0.168), (4.5e+008, 0.178), (5e+008, 0.185), (5.5e+008, 0.19), (6e+008, 0.195), (6.5e+008, 0.2), (7e+008, 0.203), (7.5e+008, 0.205)

Effect_of_Removal_Expenditure_on_Removal_Rate = GRAPH(Expenditure_on__Plastic_Removal)



(0.00, 0.00), (5e+007, 0.02), (1e+008, 0.05), (1.5e+008, 0.08), (2e+008, 0.12), (2.5e+008, 0.15), (3e+008, 0.18), (3.5e+008, 0.21), (4e+008, 0.23), (4.5e+008, 0.25), (5e+008, 0.27), (5.5e+008, 0.28), (6e+008, 0.29), (6.5e+008, 0.3), (7e+008, 0.31), (7.5e+008, 0.32), (8e+008, 0.32), (8.5e+008, 0.33), (9e+008, 0.33), (9.5e+008, 0.34), (1e+009, 0.34)

Figure 5.5. Equations of the Dynamic Model (Part Ⅳ)

6. MODEL VALIDATION

In line with the guidelines proposed by Forrester and Senge, our validation process centers on confirming both the structural soundness and behavioral credibility of the model. First, we verified that the causal loop diagram (CLD) accurately represents the relationships outlined in our problem statement and that the stock-flow diagram captures these dynamics in a way consistent with real world processes. During this structure verification step, we ensured that each equation and parameter aligns with observed knowledge of the Turkish electronics industry and its environmental impacts. We also applied dimensional consistency checks to confirm that units on both sides of our equations match appropriately, avoiding unrealistic scaling or hidden assumptions.

To further test the model's robustness, we conducted extreme conditions trials (such as assuming very high or very low VAT collections) and examined whether the resulting behavior still matched logical expectations for the real system. We found that the system handled these extremes in a stable way, indicating that our model does not produce unmanageable or nonsensical results outside normal operating ranges. Finally, we performed behavior reproduction checks to see if the overall output follows anticipated patterns (such as rising or falling plastic waste in water). These steps give us confidence that our model is suitable for examining how VAT on electronics may influence the amount of plastics in Turkish waters over the next 20 years.

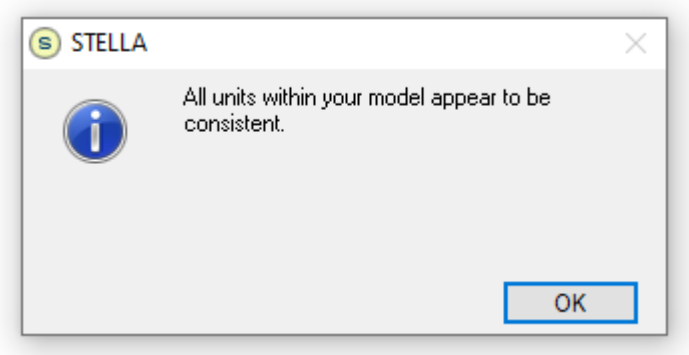


Figure 6.1. Dimensionally Consistency of the Dynamic Model

7. RESULTS

Table of Stocks

03:27 29.12.2024 Table of Critical Stocks and Variables (Table of Stocks)

Years	Amount of VAT Collected f	Money in Waste Managem	Inventory of Electronics in	Production Workers in Elec	Electronics in Use in Turkey	Amount of Recycled Plastic	Plastic Materials at Hospita	Amount of Plastics in Wate	Generated Plastics Waste i
,00	4.540.000.000,00	1.000.000.000,00	9.000.000,00	150.000,00	90.000.000,00	10.000.000,00	500.000,00	5.000.000,00	13.493.750,00
1,00	6.256.912.132,56	2.192.429.677,94	8.844.936,52	146.616,21	108.126.907,91	12.620.896,94	612.171,53	5.378.906,67	16.216.113,97
2,00	7.948.215.085,75	2.587.205.046,57	8.187.640,77	147.663,16	123.416.296,05	15.716.723,89	758.903,94	5.786.429,24	18.533.176,97
3,00	9.614.290.841,92	2.729.336.830,36	8.590.481,59	152.897,03	136.312.339,36	19.163.433,45	938.888,54	6.237.650,60	20.510.023,87
4,00	11.255.515.685,77	2.791.273.718,56	9.063.151,52	148.361,46	147.189.683,27	22.885.944,70	1.148.921,46	6.728.996,90	22.200.711,39
5,00	12.872.260.289,44	2.827.583.385,90	8.639.576,61	146.152,75	156.364.327,61	26.821.649,71	1.385.100,11	7.255.365,27	23.650.354,27
6,00	14.464.889.796,18	2.855.535.160,25	8.109.476,65	150.998,80	164.102.806,72	30.918.356,69	1.643.362,67	7.811.660,14	24.896.697,14
7,00	16.033.763.902,79	2.880.596.883,11	8.868.058,86	151.165,54	170.629.932,65	35.132.588,10	1.919.792,96	8.393.299,46	25.971.380,15
8,00	17.579.236.940,94	2.904.502.276,45	8.950.806,69	147.130,05	176.135.326,19	39.428.166,76	2.210.779,72	8.996.319,17	26.900.967,36
9,00	19.101.657.957,10	2.927.803.508,88	8.349.922,53	146.238,87	180.778.925,59	43.775.037,31	2.513.087,13	9.617.340,28	27.707.788,65
10,00	20.601.370.791,47	2.950.678.844,29	8.350.049,85	153.486,00	184.695.632,95	48.148.281,56	2.823.872,93	10.253.498,07	28.410.633,29
11,00	22.078.714.155,53	2.973.188.187,81	9.034.363,28	149.376,64	187.999.233,28	52.527.293,93	3.140.676,56	10.902.367,75	29.025.324,04
12,00	23.534.021.708,65	2.995.353.942,84	8.766.186,72	146.410,75	190.785.700,11	56.895.090,16	3.461.391,61	11.561.897,11	29.565.194,43
13,00	24.967.622.133,36	3.017.186.595,60	8.123.742,16	148.968,19	193.135.983,63	61.237.726,84	3.784.231,10	12.230.595,40	30.041.486,69
14,00	26.379.839.209,62	3.038.692.810,92	8.727.126,69	152.255,64	195.118.362,37	65.543.813,70	4.107.690,75	12.907.830,44	30.463.684,41
15,00	27.770.991.887,93	3.059.877.994,64	9.053.927,98	147.821,57	196.790.426,81	69.804.103,46	4.430.513,17	13.592.421,84	30.839.791,23
16,00	29.141.394.361,41	3.080.747.105,04	8.552.615,27	146.028,59	198.200.752,40	74.011.146,83	4.751.654,45	14.283.311,91	31.176.564,58
17,00	30.491.356.136,70	3.101.304.910,34	8.138.223,48	152.200,88	199.390.310,79	78.159.002,08	5.070.253,96	14.979.597,40	31.479.712,07
18,00	31.821.182.103,89	3.121.556.070,92	8.961.300,23	150.825,46	200.393.660,06	82.242.990,63	5.385.607,55	15.680.505,59	31.754.056,56
19,00	33.131.172.605,39	3.141.505.166,03	8.972.483,42	146.969,40	201.239.948,72	86.259.491,26	5.697.144,09	16.385.374,06	32.003.675,06
Final	34.421.623.503,75	3.161.156.702,93	8.348.467,29	146.189,30	201.953.762,45	90.205.766,75	6.004.405,08	17.093.633,73	32.232.015,69

Table 7.1. Table of Stocks and A Variable by Year

The table presents the long term evolution of stocks and a variable in our system model over 20 years. It shows how financial flows (Amount of VAT Collected, Money in the Waste Management Fund), industry stocks (Inventory of Electronics, Production Workers), consumer usage (Electronics in Use) and environmental factors (Amount of Recycled Plastics, Plastic Materials at Hospitals, Amount of Plastics in Water, Generated Plastics Waste) change together over time.

The table also reveals that even with ongoing recycling and plastic removal efforts, some level of plastic waste still accumulates in the environment. Overall, the table captures the dynamic interplay among government budgets, industrial production, consumer demand and environmental outcomes, highlighting how policy decisions and industry behavior jointly influence plastic pollution levels and tax revenues.

7.1. TIME SERIES ANALYSIS

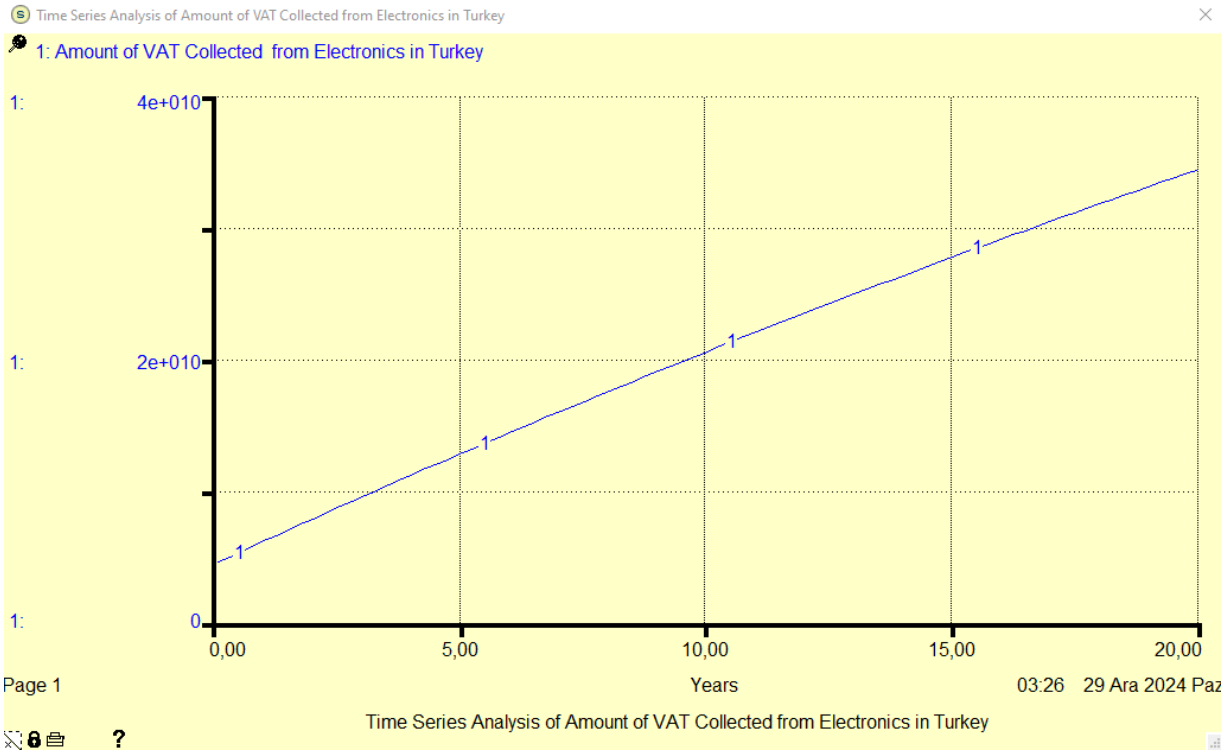


Figure 7.1. Time Series Plot of Amount of VAT Collected from Electronics in Turkey

The graph shows a steady climb in VAT revenue over 20 years, even though the model holds the average price and overall sales relatively constant. Because demand is inelastic, higher or stable prices do not drastically cut into the number of devices sold, so VAT income keeps accumulating. At the same time, part of the collected tax flows out as Government Funding for waste management. However, that outflow does not significantly slow the upward trend in total VAT collected. Overall, this reflects a situation where stable sales volumes, consistent prices and inelastic demand enable continuous growth in accumulations of the taxes, despite ongoing government expenditures.

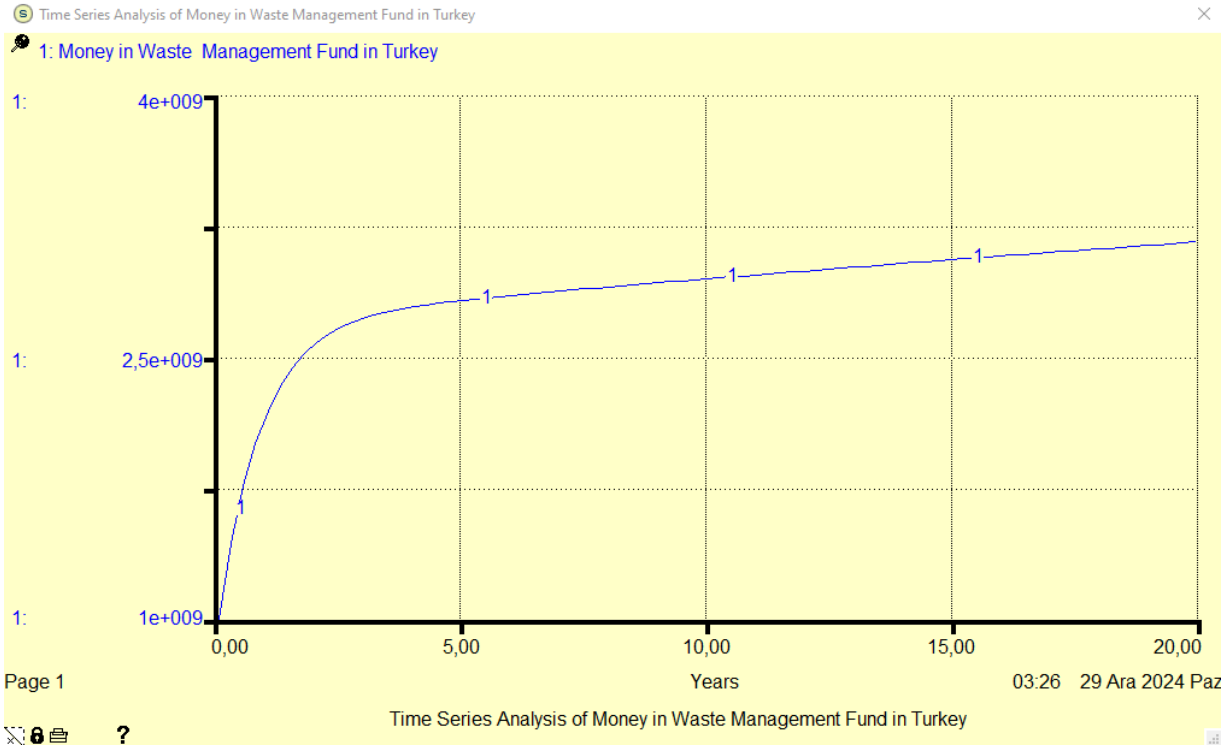


Figure 7.2. Time Series Plot of Money in Waste Management Fund in Turkey

The graph illustrates the evolution of the Waste Management Fund in Turkey over a 20 year period. In the early years, the fund experiences a rapid increase, largely due to the initial allocation of government resources derived from VAT revenue. As time progresses, the growth rate begins to level off, indicating that although expenditures on recycling and waste disposal continue, they do not outpace the steady inflow of tax derived funding. Eventually, the curve settles into a slower but still upward trajectory, suggesting a near equilibrium between incoming funds and ongoing expenditures. Overall, the fund's balance continues to grow, reflecting a system in which consistent financing (despite continuous spending) maintains a positive long term trend.

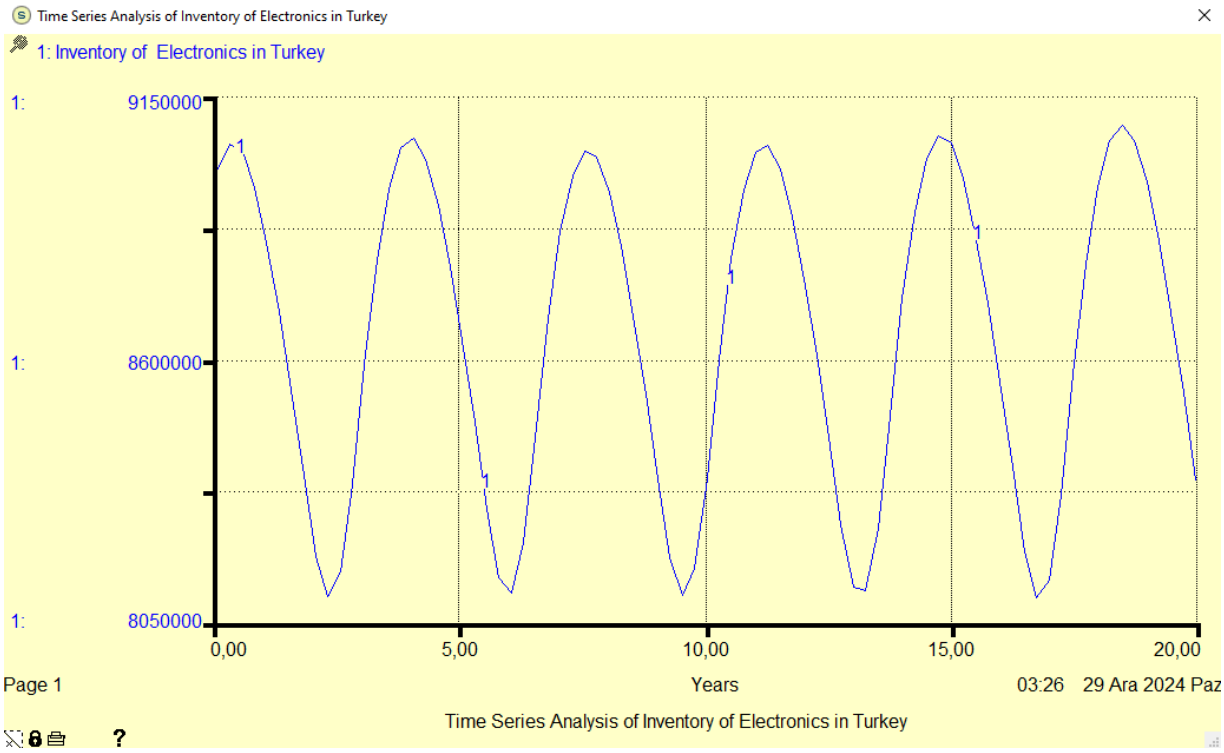


Figure 7.3. Time Series Plot of Inventory of Electronics in Turkey

The graph reveals cyclical fluctuations in Turkey's electronic device inventory over a 20 year period. The recurrent peaks represent times when supply accumulates faster than it is consumed (often following production boosts or market releases) while the troughs reflect periods of higher demand or usage depleting the available stock. The relatively consistent amplitude of these waves suggests stable underlying factors, such as predictable production rates and consumer behavior patterns. Overall, the inventory repeatedly returns to comparable high and low points, indicating a cyclical yet balanced market dynamic in which surges in supply are regularly followed by periods of increased demand.

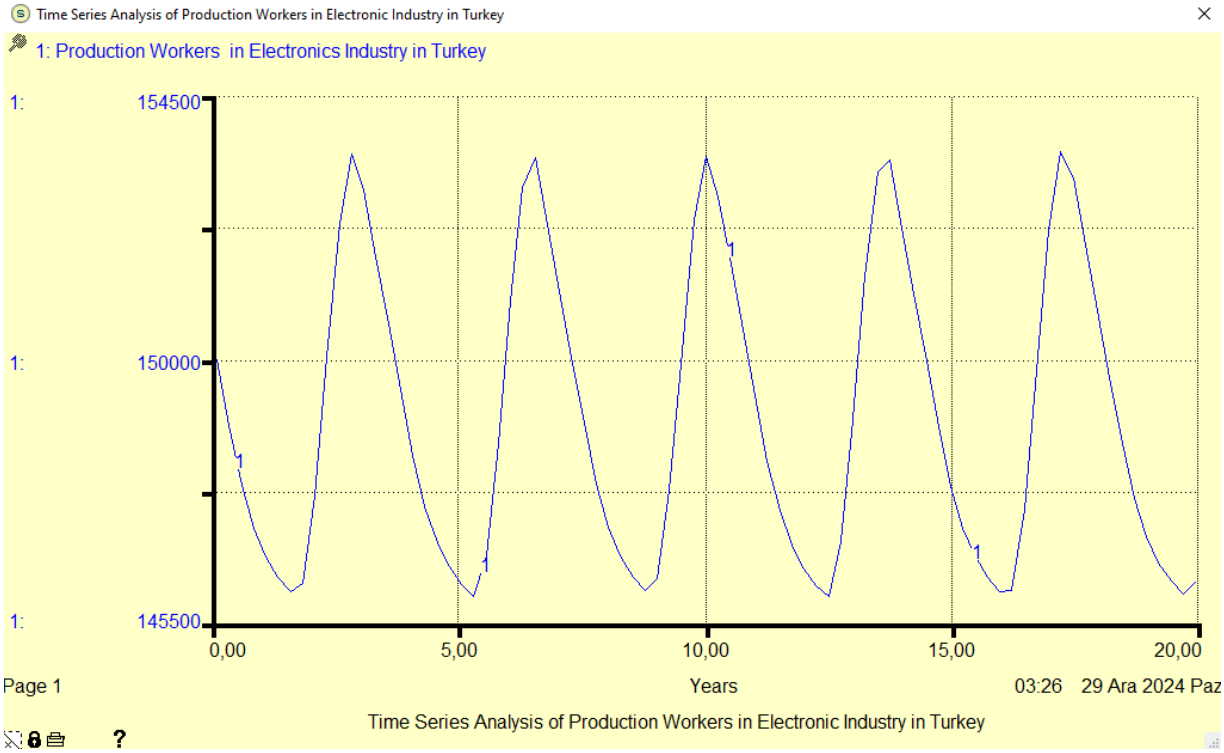


Figure 7.4. Time Series Plot of Production Workers in Electronics Industry in Turkey

The graph depicts cyclical fluctuations in the number of production workers in Turkey's electronics industry over a 20 year span. Each peak represents a surge in production activity (often driven by product launches or increased market demand) while each trough coincides with slower periods or workforce adjustments. The regularity of these oscillations suggests that the industry's labor requirements follow a fairly predictable pattern. Overall, the amplitude remains relatively steady, indicating that expansions and contractions in workforce needs tend to balance out over time, creating a repeated cycle of highs and lows in employment levels.

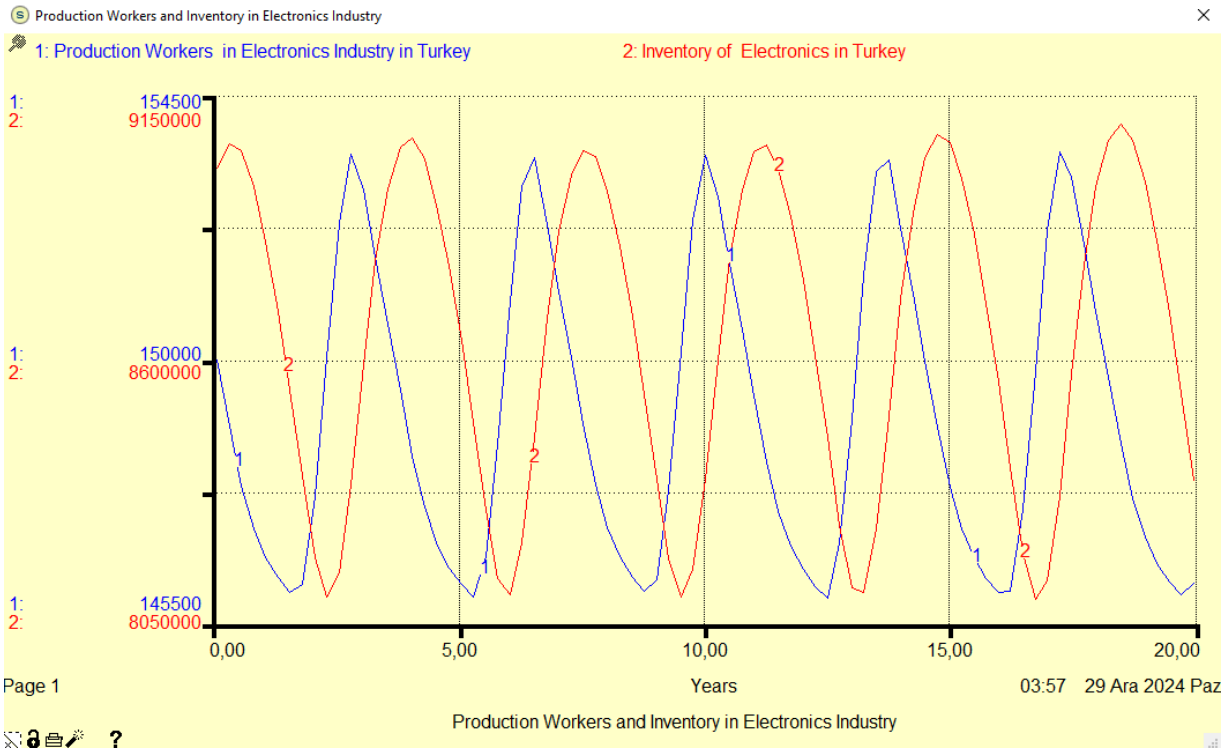


Figure 7.5. Time Series Plot of Production Workers and Inventory in Electronics

This composite time series plot shows two cyclical patterns over a 20 year span: the number of production workers (blue line) and the inventory of electronic devices (red line) in Turkey's electronics industry. Both curves exhibit recurring peaks and troughs, but they are slightly offset from each other.

When the production workforce is at a peak, it often indicates an upcoming rise in output, which can then translate into higher inventory if demand does not immediately absorb all newly produced units. Conversely, when workforce numbers dip, it may coincide with or soon be followed by an inventory decline, reflecting reduced production activity. The close correlation between these two cycles underscores how labor availability and manufacturing output jointly shape the supply side of the market.

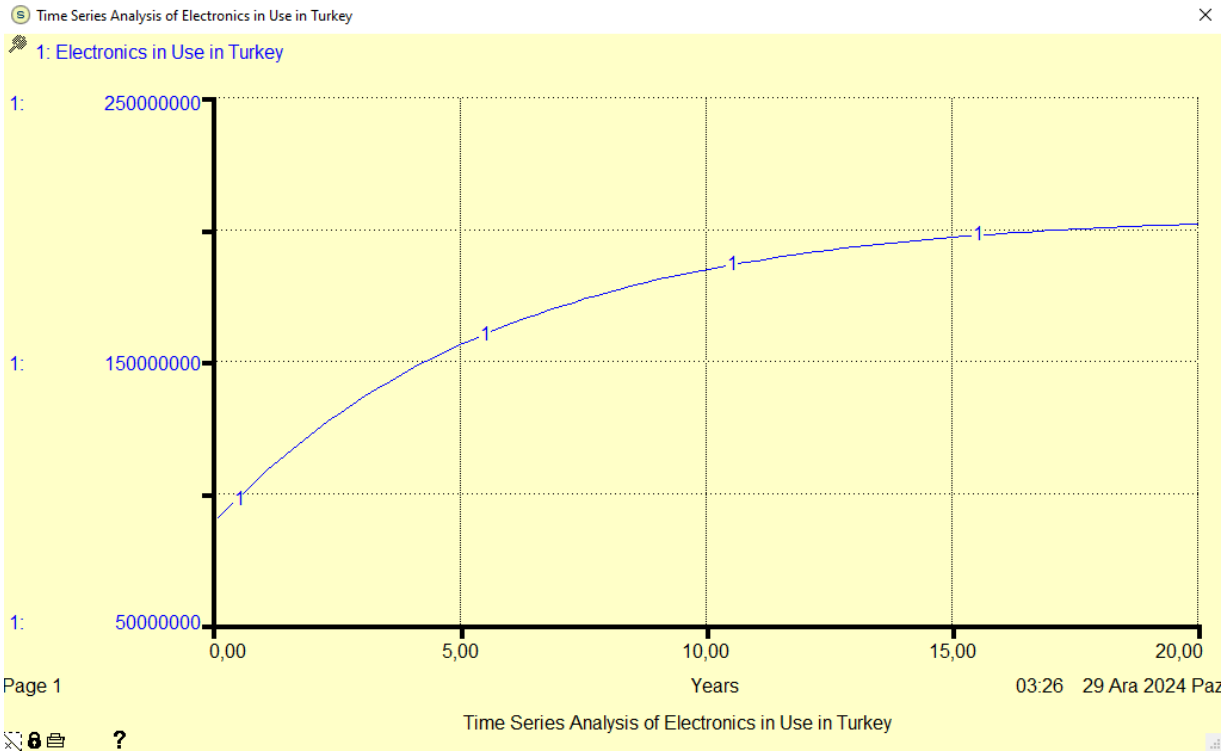


Figure 7.6. Time Series Plot of Electronics in Use in Turkey

The graph illustrates the growth in the number of electronic devices in use in Turkey over a 20 year period. The curve shows a rapid increase in the earlier years, reflecting widespread adoption and expanding usage of electronics. As the years progress, the growth rate begins to slow down, suggesting a saturation point where the majority of the population already has access to the necessary electronic devices. The gradual flattening of the curve indicates a stabilization in the market, likely due to a balance between new purchases and the abandonment of older devices.

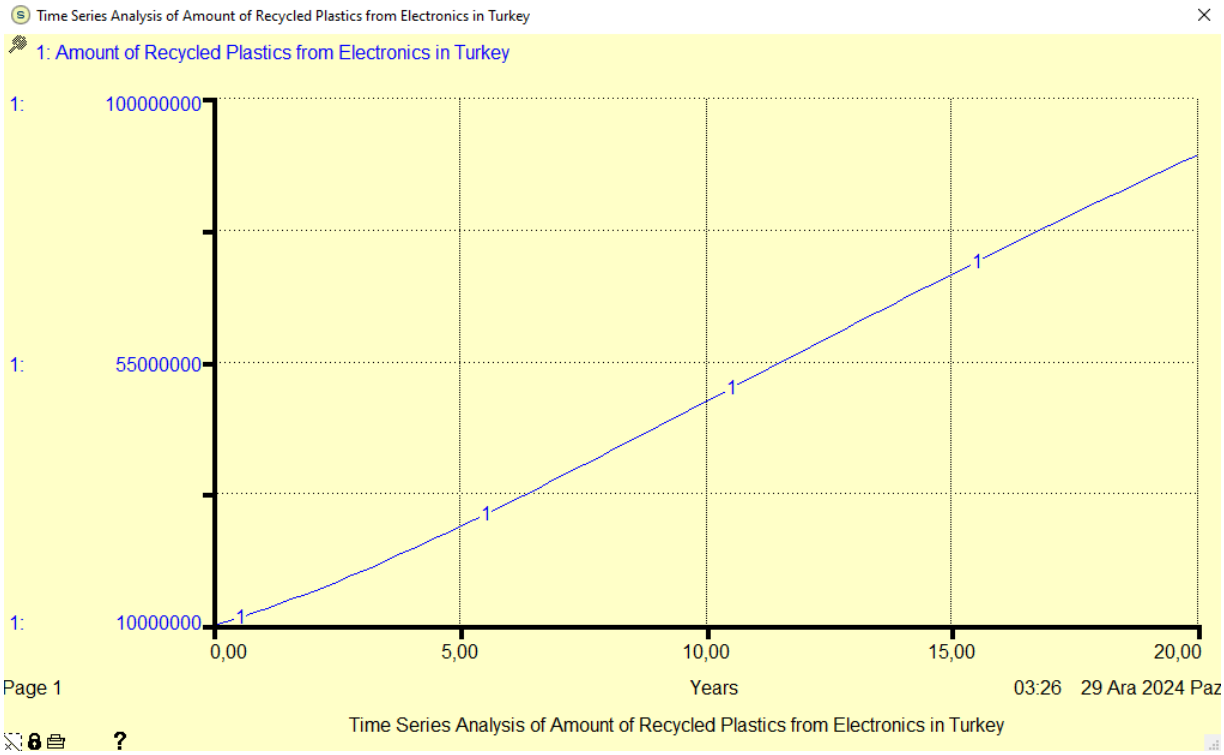


Figure 7.7. Time Series Plot of Amount of Recycled Plastics from Electronics

The graph represents the growth in the amount of recycled plastics obtained from electronic devices in Turkey over a 20 year period. The steady upward trend indicates a consistent improvement in recycling efforts, driven by advancements in waste management systems and increased awareness of environmental sustainability. The linear growth pattern suggests that recycling capacity has expanded proportionally over time, likely supported by government initiatives and industry practices. Overall, the graph highlights the progress made in minimizing plastic waste through effective recycling processes, which contribute to reducing the environmental footprint of electronic products.

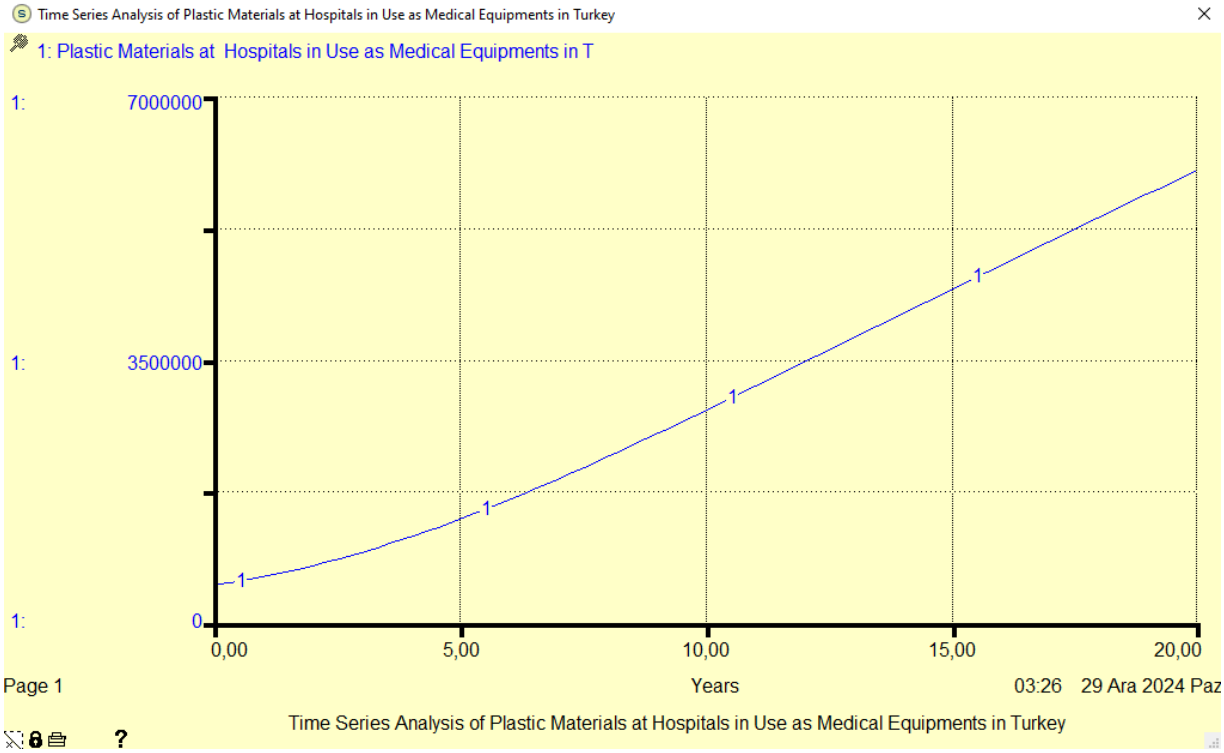


Figure 7.8. Time Series Plot of Plastic Materials at Hospitals in Use in Turkey

The graph demonstrates the increasing use of plastic materials in hospitals as medical equipment in Turkey over a 20 year period. The curve shows a steady and consistent upward trend, indicating a growing reliance on plastic based medical tools and devices in healthcare facilities. This rise can be attributed to advancements in medical technology, increased demand for single use, disposable or durable medical equipment and the expanded capacity of hospitals to meet the needs of a growing population. The linear nature of the growth suggests that the adoption of plastic materials in the medical field is progressing at a stable rate, driven by both practical and hygienic considerations in modern healthcare systems.

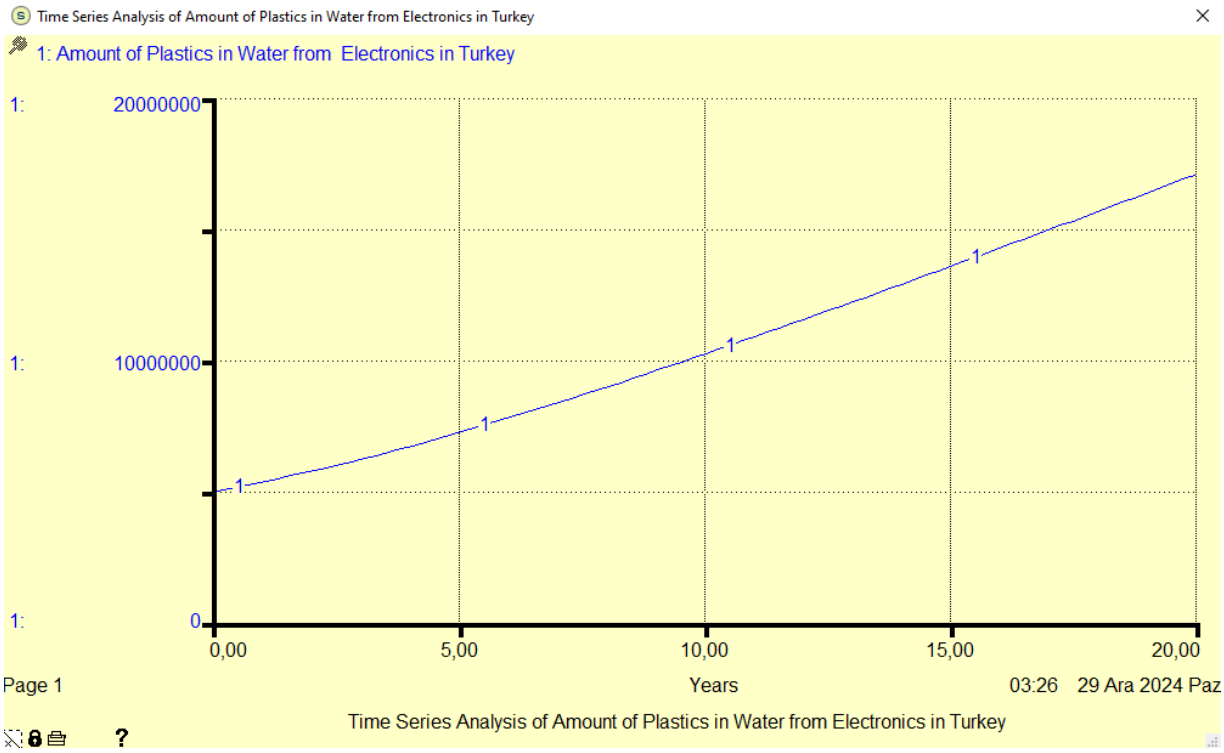


Figure 7.9. Time Series Plot of Amount of Plastics in Water from Electronics

The graph illustrates the increasing amount of plastics in water derived from electronic devices in Turkey over a 20 year period. The upward trend indicates that despite efforts in recycling and waste management, a significant portion of plastic waste from electronics still ends up in water systems. This consistent growth can be attributed to the rising use of electronics.

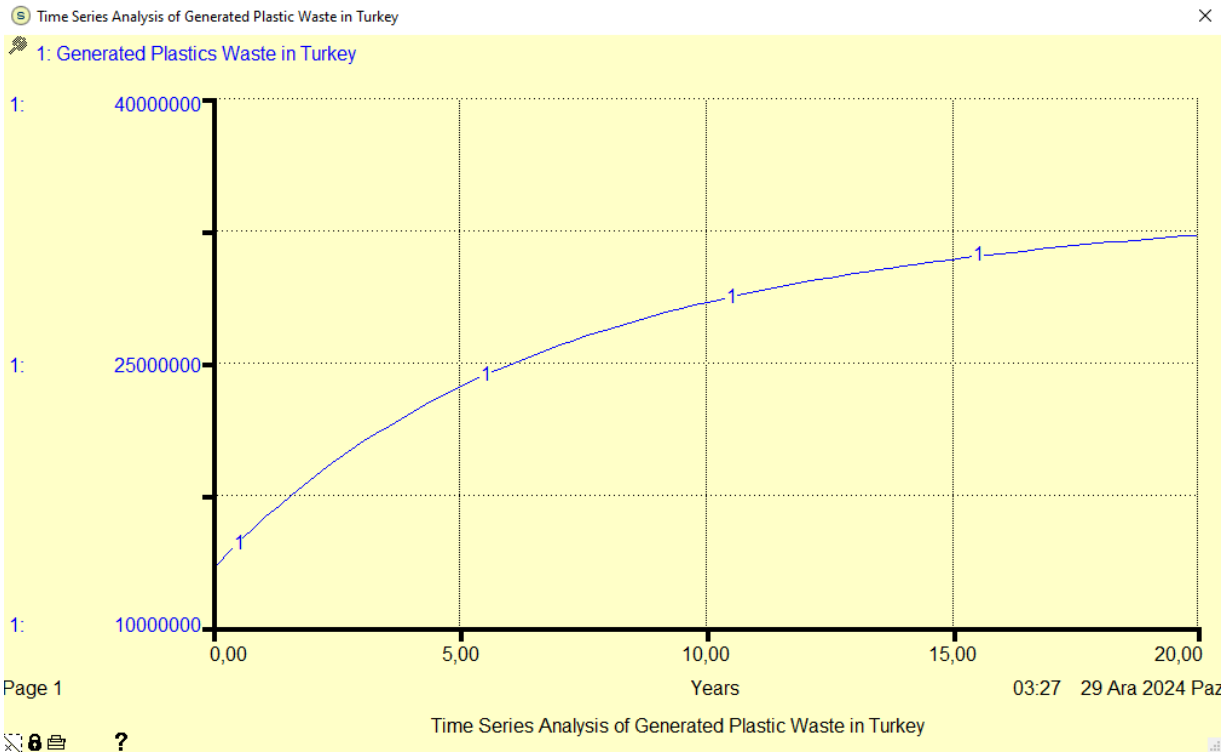


Figure 7.10. Time Series Plot of Generated Plastic Waste from Electronics

The graph depicts the generation of plastic waste in Turkey over a 20 year period. The curve shows a rapid increase in the initial years, reflecting the growing consumption of plastic materials, particularly in industries such as electronics. As the timeline progresses, the growth rate slows down, indicating a stabilization as waste management practices and recycling efforts begin to mitigate the rate of increase.

7.2. SCATTER PLOT ANALYSIS

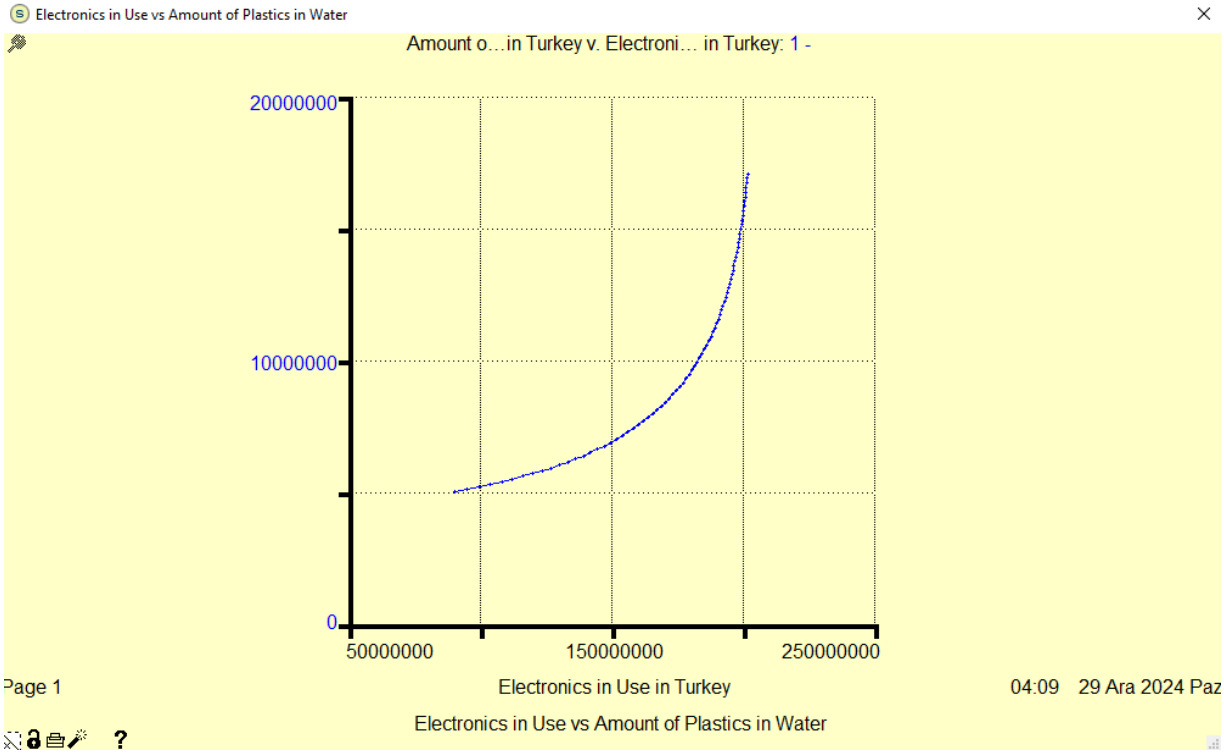


Figure 7.11. Scatter Plot of Electronics in Use vs Amount of Plastics in Water

The plot illustrates the relationship between the number of electronics in use and the amount of plastics in water in Turkey. The graph shows a significant upward trend, indicating that as the usage of electronic devices increases, the amount of plastic pollution in water also grows. This highlights the impact of disposal or recycling of electronics on environmental plastic waste. The graph underscores the importance of enhancing recycling systems and raising public awareness to address the environmental challenges associated with increasing electronics usage.

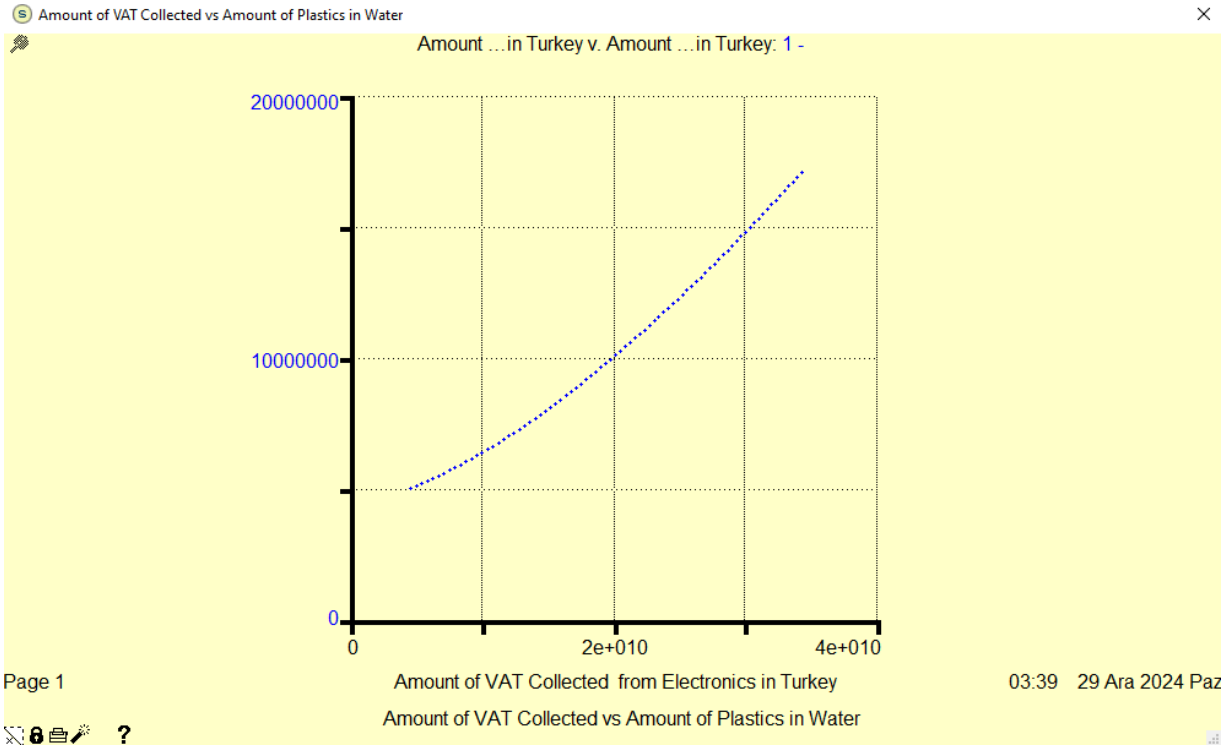


Figure 7.12. Scatter Plot of Amount of VAT Collected vs Amount of Plastics in Water

The plot represents the relationship between the amount of VAT collected from electronics and the amount of plastics found in water in Turkey. The positive correlation shown by the upward trend suggests that as VAT revenue from electronics increases, the associated plastic waste in water also rises. This relationship highlights the indirect impact of increased electronics consumption (reflected in higher VAT collection) on environmental plastic pollution. The graph emphasizes the importance of integrating sustainable waste management practices with policies related to electronics production and taxation to mitigate environmental harm.

7.3. SENSITIVITY ANALYSIS

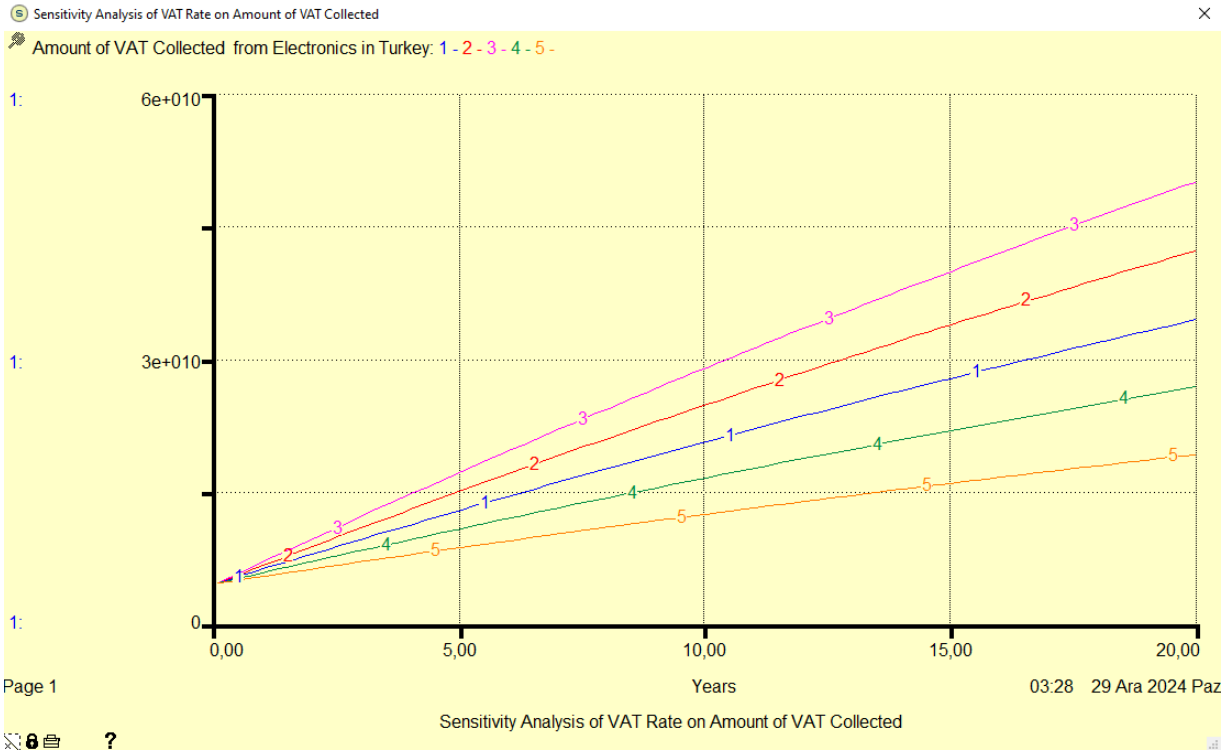


Figure 7.13. Sensitivity Analysis of VAT Rate on Amount of VAT Collected

The graph illustrates the sensitivity analysis of VAT rates on the total amount of VAT collected from electronics in Turkey over a 20 year period. Each curve represents a run on different VAT rates: 1 (20%), 2 (22%), 3 (24%), 4 (18%) and 5 (16%). The graph demonstrates a clear positive relationship: as the VAT rate increases, the total VAT revenue collected also grows significantly over time due to the low price elasticity of electronics (between 0 and 1 at around the market price). For example, 3 (24%) shows the steepest growth trajectory, generating the highest revenue, while 5 (16%) represents the lowest revenue collected. The separation between the curves widens as the years progress, highlighting the compounding effect of higher VAT rates on revenue accumulation. This analysis underscores the critical role of VAT rates in shaping government tax revenues and provides valuable insights for fiscal policy decisions.

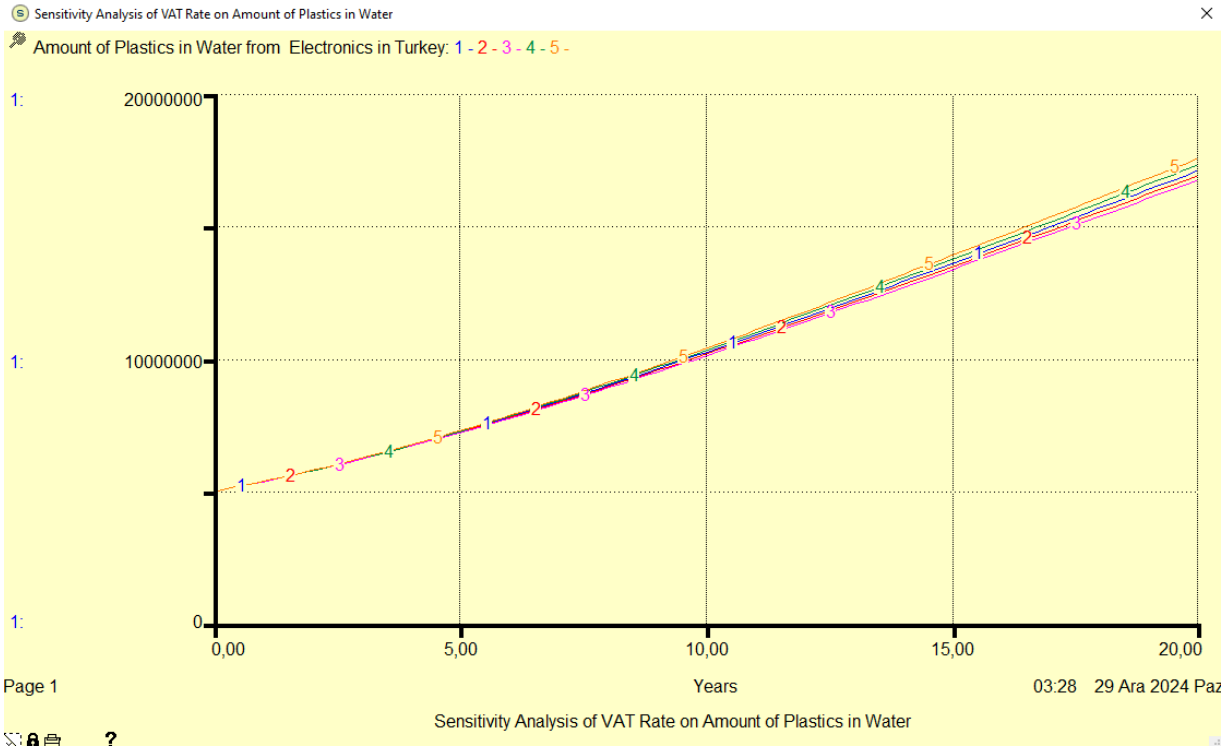


Figure 7.14. Sensitivity Analysis of VAT Rate on Amount of Plastics in Water

The graph demonstrates the sensitivity of the Amount of Plastics in Water to different VAT rates applied to electronics over a 20 year period. Each curve represents a run on different VAT rates: 1 (20%), 2 (22%), 3 (24%), 4 (18%) and 5 (16%). Although higher VAT rates (such as 3 at 24%) lead to slightly reduced sales due to the low price elasticity of electronics (between 0 and 1 at around the market price), the dominant factor remains the number of electronics in use, which drives the system's dynamics. As a result, variations in VAT rates have a limited impact on reducing plastic pollution. The overall upward trend by time across all scenarios underscores the importance of integrating effective waste management and recycling policies to address plastic accumulation, as fiscal measures alone are insufficient to mitigate environmental harm.

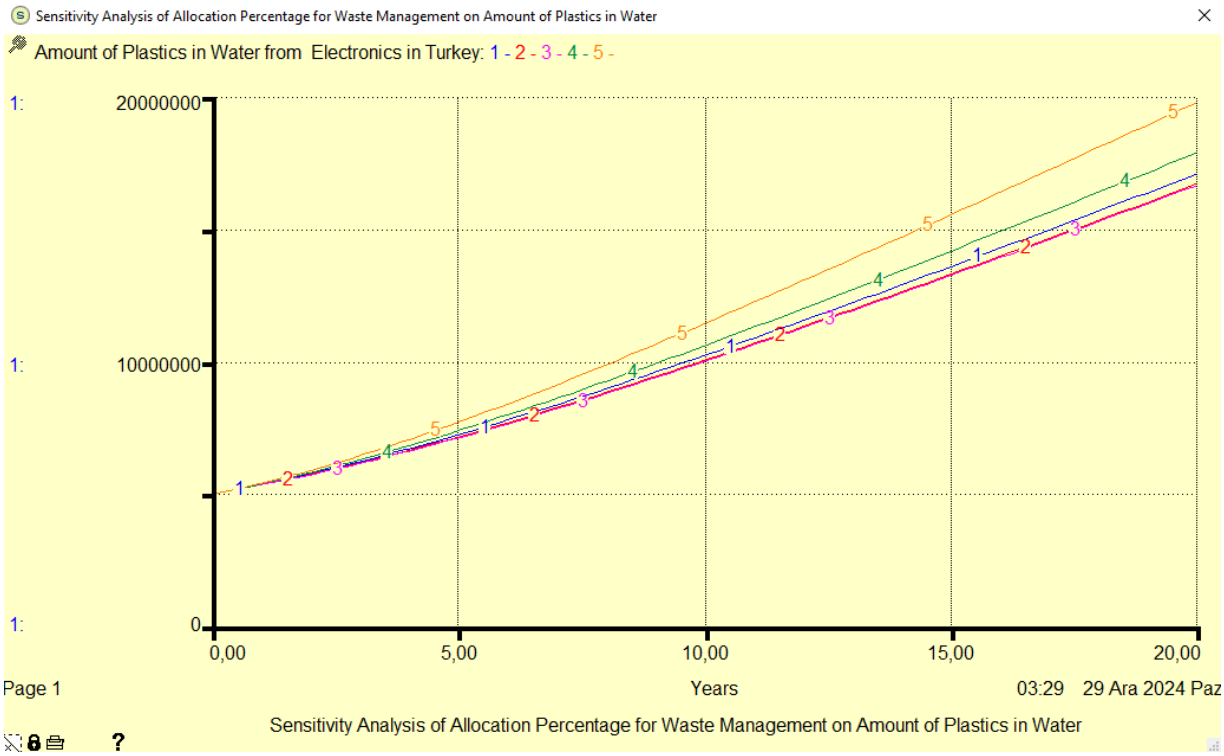


Figure 7.15. Sensitivity Analysis of Allocation Percentage on Amount of Plastics in Water

The graph presents a sensitivity analysis of the allocation percentage for waste management on the amount of plastics in water derived from electronics in Turkey over a 20 year period. Each curve represents a run on different allocation percentages: 1 (1.5%), 2 (2%), 3 (2.5%), 4 (1%) and 5 (0.5%). Higher allocation percentages, such as 3 (2.5%), correspond to lower amounts of plastics in water, indicating a significant reduction in plastic pollution. Conversely, lower allocation percentages, such as 5 (0.5%), result in higher levels of plastic pollution over time. The divergence between the curves demonstrates that increasing the allocation for waste management leads to a noticeable decrease in environmental plastic waste. This analysis underscores the importance of directing more resources toward waste management and recycling efforts to mitigate the environmental impact of electronic waste effectively and achieve sustainable waste management goals.

8. CONCLUSION

This study effectively uses system dynamics modelling to analyse the impact of VAT on the amount of plastic in water in Turkey's electronics sector. The application of Causal Loop Diagrams (CLDs) and Stock-Flow Diagrams generated using STELLA software has provided a comprehensive understanding of the complex feedback mechanisms inherent in this sector.

One of the key findings of this study is the impact of VAT collection on the government's waste management budget and its subsequent impact on the amount of plastics in water from electronics in Turkey. As VAT collection increases due to higher sales, expenditures on recycling and plastic removal programs also increase. However, while these expenditures are necessary, they lead to secondary waste streams, particularly from recycled materials reused in medical equipment at hospitals, which ultimately re-enter the waste cycle. Also, the increase in electronics sales increases plastic waste through direct disposal of end-of-life products.

The impact of changes in VAT rates on the dynamic model is assessed through sensitivity analysis. Considering the inelastic nature of price around market price, reductions in VAT rates increase sales by reducing average product prices. This, in turn, leads to an increase in the amount of plastic waste. However, lower VAT revenues limit the resources available for recycling and plastic removal programs, negatively affecting the accumulation of plastics in water. This interaction is further confirmed by sensitivity analyses of the percentage of allocation, which underlines the critical role of adequate financing in reducing plastic waste. This study highlights the necessity of comprehensive waste management strategies to effectively address the problem of plastic pollution in water and VAT collection as an essential financial instrument for fiscal policies.

Furthermore, this study highlights the importance of system dynamics and modelling in capturing and simulating real-world complexities in understanding complex systems. The comprehensive representation of interconnected variables and dynamic relationships makes the model a tool to provide strategic insights to policy makers and industry stakeholders.

In conclusion, these findings emphasise that VAT collection, plastic removal and recycling efforts play a vital role in addressing the impact of electronics on the amount of plastics in water. Integrating fiscal policies with effective waste management strategies offers the potential to create a more sustainable electronics industry and reduce plastic waste in water. The insights from the study could guide policy makers in achieving financial and environmental sustainability goals, ultimately affecting both the industry and the environment positively.

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